COMPUTER NETWORKS

MODULE 1 – Introduction, Application Layer & Transport Layer

KTU B.Tech | CSBS | 2024 Scheme | S5

Study Notes for Examination Preparation

Prepared according to the Module 1 syllabus supplied by the student. Explanations are simplified for revision. Verify protocol details and practical requirements with class materials and the prescribed textbook.

Module 1 topics: Internet overview, protocol layering, application-layer paradigms, WWW, HTTP, FTP, email, DNS, P2P and BitTorrent, transport-layer services, UDP, TCP, and socket programming introduction.

1. Introduction to Computer Networks

Computer Network: A computer network is a group of two or more computers or devices connected through communication links so that they can exchange data and share resources.

Examples

- The Internet connecting computers around the world.
- A college Wi-Fi network connecting student devices.
- An office network sharing printers, files and applications.
- An ATM network connecting bank machines to bank servers.

Advantages of Computer Networks

- Resource sharing – devices, software and files can be shared.
- Communication – supports email, messaging and video calls.
- Data sharing – users can access common information.
- Cost reduction – shared resources reduce duplication.
- Centralized management – data and access can be managed centrally.

2. Network Protocols

A **protocol** is a set of rules that governs communication between devices. It defines how data is formatted, transmitted and received.

Three basic elements

Element	Meaning
Syntax	The structure or format of the data.
Semantics	The meaning of each field and the action to be taken.
Timing	When data should be sent and at what rate.

Examples: HTTP, FTP, TCP, UDP and IP.

3. Protocol Layering

Network communication is divided into layers. Each layer performs a specific task, uses the services of the layer below it and provides services to the layer above it. Layering makes network design easier to understand, implement and maintain.

OSI Reference Model – Seven Layers

Layer	Name	Main function
7	Application	Provides network services to user applications.
6	Presentation	Data translation, formatting, compression and encryption.
5	Session	Establishes, manages and terminates sessions.
4	Transport	Process-to-process delivery, reliability and flow control.
3	Network	Logical addressing and routing of packets.
2	Data Link	Framing, link-level addressing and error detection.

Layer	Name	Main function
1	Physical	Transmits raw bits over a physical medium.

Mnemonic: All People Seem To Need Data Processing.

Why layering is useful

- Reduces complexity by dividing communication into manageable tasks.
- Allows one layer to change without redesigning the entire network, provided interfaces remain compatible.
- Encourages standardization and interoperability.

4. Application Layer

The application layer provides network services to applications used by people, such as web browsers, email clients and file-transfer programs.

Application-layer paradigms

Client–Server Model

- A client requests a service or resource.
- A server waits for requests and provides the requested service.
- The server is generally reachable at a known address and is often continuously available.
- Example: a web browser requesting a page from a web server.

Peer-to-Peer (P2P) Model

- Peers communicate and share resources directly.
- A device can act as both a client and a server.
- There may be no permanently required central server for transferring the shared content.
- Example: BitTorrent file sharing.

5. World Wide Web (WWW)

The **World Wide Web** is a system of interlinked web pages and resources that are accessed over the Internet using web protocols.

- Web browser – client software used to access web resources.
- Web server – stores or generates resources and sends responses.
- URL – address used to identify a resource.
- HTTP/HTTPS – protocols used to transfer web resources.
- HTML – markup language commonly used to structure web pages.

Important distinction: The Internet is the global network infrastructure; the Web is one service that operates over the Internet.

6. HTTP – Hypertext Transfer Protocol

HTTP is an application-layer protocol used for communication between web clients and servers. It follows a request–response pattern.

Basic working

- The user enters a URL or selects a link.

- The browser obtains the server's address, commonly using DNS.
- The browser sends an HTTP request to the server.
- The server processes the request and sends an HTTP response.
- The browser interprets the response and displays the content.

Common HTTP methods

Method	Purpose
GET	Requests or retrieves a resource.
POST	Submits data to be processed, often creating or triggering a resource.
PUT	Creates or replaces the state of a resource at a specified URI.
DELETE	Requests deletion of a specified resource.

HTTP and HTTPS

HTTP transfers web messages. HTTPS is HTTP communication protected using TLS, which provides encryption and helps authenticate the server. HTTPS is preferred for sensitive information.

7. FTP – File Transfer Protocol

FTP is an application-layer protocol designed for transferring files between a client and a server over a TCP/IP network.

Common operations

- Uploading files from a client to a server.
- Downloading files from a server to a client.
- Listing and navigating directories.
- Renaming and deleting files, subject to permissions.

Traditional FTP uses separate control and data connections. Because traditional FTP does not encrypt credentials and data, secure alternatives such as SFTP or FTPS are commonly preferred where security is required.

8. Electronic Mail (Email)

Email enables users to exchange electronic messages. A typical email system contains user agents, mail servers and message-transfer/retrieval protocols.

Main components and protocols

Component / Protocol	Role
User Agent	Software used to compose, send, read and manage email.
Mail Server	Stores mailboxes and transfers messages between servers.
SMTP	Primarily used to send email and transfer it between mail servers.
POP3	Retrieves messages, commonly by downloading them to a client.
IMAP	Accesses and synchronizes messages while they remain managed on the server.

Exam point: SMTP is mainly for sending; POP3 and IMAP are used for retrieving or accessing email.

9. DNS – Domain Name System

DNS is a distributed naming system that translates human-readable domain names into IP addresses and can also provide other information about domain names.

Example: A user enters *www.example.com*. DNS helps the computer find the IP address associated with that name.

Simplified DNS resolution

- The application checks available local or browser DNS cache.
- If no usable cached answer exists, the device asks a DNS resolver.
- The resolver obtains the required DNS information, possibly by querying other DNS servers.
- The resolver returns the answer to the device and may cache it.
- The browser can then contact the destination IP address.

DNS is often described as the phonebook of the Internet. Common DNS record types include A (IPv4 address), AAAA (IPv6 address), MX (mail exchanger) and CNAME (alias).

10. Peer-to-Peer Networks and BitTorrent

A P2P network allows participating computers, called peers, to share data or services directly. A peer may download and upload content.

Advantages

- Can distribute workload across many peers.
- Can reduce dependence on a single content server.
- Can scale well when many peers contribute resources.

BitTorrent

BitTorrent is a P2P file-distribution protocol. A file is divided into pieces, and a peer can obtain different pieces from multiple peers while also uploading pieces it already has.

- Seeder – a peer that has the complete file and makes pieces available.
- Leecher – a peer that is still downloading pieces; it may also upload pieces it has received.

The availability and speed of a download depend on factors such as the number of peers, their upload capacity and which pieces they possess.

11. Transport Layer

The transport layer provides logical communication between application processes running on different hosts. It uses port numbers to identify application processes.

Main functions

- Process-to-process delivery.
- Segmentation of application data and reassembly at the destination.
- Multiplexing and demultiplexing using port numbers.
- Error detection and, depending on the protocol, recovery.
- Flow control and congestion control where supported.
- Connection establishment and termination for connection-oriented communication.

12. TCP and UDP

TCP – Transmission Control Protocol

- Connection-oriented: a connection is established before normal data transfer.
- Reliable, ordered byte-stream delivery using acknowledgements and retransmissions.
- Provides flow control and congestion control.
- Used by applications that need reliable ordered delivery, such as many web, email and file-transfer services.

UDP – User Datagram Protocol

- Connectionless: no connection handshake is required before sending datagrams.
- Provides best-effort delivery; it does not guarantee delivery, ordering or duplicate protection.
- Has a small header and low protocol overhead.
- Used where low latency or application-specific control is important, such as many real-time applications and DNS queries.

Feature	TCP	UDP
Communication	Connection-oriented	Connectionless
Reliability	Reliable, with acknowledgements and retransmission	Best effort; no delivery guarantee
Ordering	Ordered byte stream	No ordering guarantee
Overhead	Higher	Lower
Typical uses	Web, email, file transfer	DNS, real-time media, online games

13. TCP Three-Way Handshake

TCP uses a three-way handshake to establish a connection and synchronize sequence numbers.

Step	Message	Meaning
1	SYN	Client requests a connection and sends its initial sequence number.
2	SYN-ACK	Server acknowledges the client's SYN and sends its own SYN.
3	ACK	Client acknowledges the server's SYN. The connection is established.

Flow: Client --SYN--> Server; Server --SYN-ACK--> Client; Client --ACK--> Server.

14. TCP Connection Termination

A common TCP connection close uses a four-segment exchange because each direction of a full-duplex connection is closed independently.

- FIN – one endpoint indicates that it has finished sending data.
- ACK – the other endpoint acknowledges the FIN.
- FIN – the other endpoint later sends its own FIN when ready.
- ACK – the first endpoint acknowledges it.

The exact close sequence can vary depending on which endpoint closes first and whether data remains to be sent.

15. UDP Communication

UDP sends independent datagrams without first establishing a transport connection. The application sends a datagram to a destination IP address and port.

Advantages and limitations

- Advantages: small header, low overhead and no connection setup delay.
- Limitations: no built-in retransmission, ordering, flow control or congestion control.
- If reliability is needed, the application may implement suitable mechanisms itself.

16. Socket Programming – Syllabus Practical Topics

A **socket** is a software endpoint used by an application to send or receive data over a network. A socket is commonly identified by an IP address, transport protocol and port number.

Topic	Meaning
TCP socket	Socket used for reliable, connection-oriented TCP communication.
TCP client	Program that initiates a connection to a server.
TCP server	Program that listens for and accepts client connections.
UDP socket	Socket used to send and receive UDP datagrams.
I/O multiplexing	Technique for monitoring multiple I/O sources in one program.
select()	System call/interface that waits for activity on a set of file descriptors.
poll()	Interface that monitors multiple file descriptors for events.
Advanced I/O	More advanced I/O techniques beyond basic blocking read/write operations.

Basic TCP client–server sequence

- Server creates a socket.
- Server binds the socket to an address and port.
- Server listens for incoming connections.
- Client creates a socket and connects to the server.
- Server accepts the connection.
- Client and server exchange data.
- The sockets are closed when communication is complete.

Basic UDP communication

- Sender creates a UDP socket.
- Sender transmits a datagram to the receiver's IP address and port.
- Receiver binds a UDP socket to its local port and receives the datagram.
- No TCP-style connection establishment is required.

17. Important Examination Questions

Short-answer / 3-mark practice

- Define a computer network and state any two advantages.
- What is a network protocol? Explain syntax, semantics and timing.
- What is protocol layering?
- List the seven layers of the OSI model with their functions.

- Define HTTP. Name any two HTTP methods.
- Differentiate SMTP, POP3 and IMAP.
- What is DNS? Why is it needed?
- What is a P2P network?
- Define seeder and leecher in BitTorrent.
- What is a socket?
- Differentiate TCP and UDP.
- What is the TCP three-way handshake?

Long-answer / 9-mark practice

- Explain the OSI reference model with a neat layered diagram.
- Explain HTTP request–response communication and common HTTP methods.
- Explain DNS and the steps involved in name resolution.
- Describe the components of an email system and explain SMTP, POP3 and IMAP.
- Explain P2P networks and the working of BitTorrent.
- Compare TCP and UDP in detail with suitable applications.
- Explain TCP connection establishment using the three-way handshake and connection termination.
- Explain transport-layer services and the basic sequence of TCP client–server socket communication.

18. Quick Revision Sheet

Topic	Remember
Protocol	Rules for communication: syntax, semantics, timing.
OSI	7 layers: Application, Presentation, Session, Transport, Network, Data Link, Physical.
HTTP	Web request–response protocol.
FTP	File transfer protocol.
SMTP	Sending email.
POP3 / IMAP	Retrieving or accessing email.
DNS	Domain name to IP address resolution.
P2P	Peers share resources directly.
TCP	Connection-oriented, reliable and ordered.
UDP	Connectionless, low overhead, best effort.
Handshake	SYN → SYN-ACK → ACK.
Socket	Application endpoint for network communication.

Study Note

These notes are a simplified study aid based on the Module 1 syllabus provided. They are not an official KTU publication and do not guarantee the exact wording or selection of examination questions. Use the prescribed textbook, class notes and practical-lab instructions for complete preparation.